WellMeadows Hospital

Management System

Project Documentation

IT223 - Web Systems & Technologies

MEMBERS:

ANASCO, KYLE CHRISLER

CASTILLO, JOHN BENEDICT

PARCO, EDJILL NINO

REBUCAS, RENXELLE DAVE

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1. Introduction

WellMeadows Hospital Management System is a web-based application developed to digitize and streamline the core administrative and clinical operations of WellMeadows Hospital. The system replaces paper-based and manual processes with an integrated platform that manages patient records, staff coordination, ward and bed allocation, appointment scheduling, treatment documentation, and billing.

The system is built on the Laravel 13 PHP framework with a PostgreSQL relational database backend. Business logic is encapsulated in PostgreSQL stored functions, which are called from Laravel controllers using raw SQL queries. Access to all system features is governed by Role-Based Access Control (RBAC), ensuring that each user type sees only the functionality relevant to their responsibilities.

The application was developed as an academic project modeled after the WellMeadows Hospital case study, with the objective of demonstrating a fully functional, multi-module hospital information system using modern web technologies.

2. Objectives

General Objective

To design and develop a web-based hospital management system that centralizes and automates the key operational workflows of WellMeadows Hospital, improving efficiency, data accuracy, and information access for hospital staff.

Specific Objectives

- To implement a Patient Management module that allows registration, updating, and viewing of patient demographic and medical information, including admission and discharge tracking.

- To develop a Staff and Department Management module that manages staff records, qualifications, work experience, ward assignments, and shift scheduling.
- To build a Ward and Bed Management module that tracks ward details, bed availability, bed allocation to patients, and overall ward occupancy.
- To create an Appointment and Treatment module that handles scheduling of patient appointments, recording of diagnoses and treatments, and assignment of clinical staff to patient treatments.
- To produce a Billing and Reporting module that generates itemized patient bills, records payments, tracks payment status, and provides financial summary reports for administrators.
- To enforce Role-Based Access Control (RBAC) so that Administrators, Receptionists, and Charge Nurses each access only the features appropriate to their role.

3. Scope and Limitations

Scope

The system covers the following functional areas:

- **Patient Management** — Registration, profile updates, medical record viewing, medication monitoring, and admission/discharge tracking.
- **Staff Management** — Staff registration and profiling, qualification and work experience recording, ward assignment, and rota/schedule management.
- **Ward and Bed Management** — Ward creation and maintenance, bed inventory, bed assignment to admitted patients, and bed availability monitoring.
- **Appointment and Treatment** — Outpatient appointment scheduling, appointment status management, diagnosis recording, treatment documentation, and staff-to-treatment assignment.
- **Billing and Reporting** — Bill generation with line items, payment recording (partial and full), bill cancellation, and financial reporting by status.
- **Authentication and RBAC** — Secure login with email/password, role-based dashboard redirection, and middleware-protected routes per user role.

Limitations

- The system does not integrate with external medical devices or laboratory information systems.
- There is no real-time notification system (e.g., alerts for bed availability or appointment reminders).
- The billing module does not connect to insurance provider APIs for automated claims verification.
- The system does not support multi-hospital or multi-branch configurations — it is scoped to a single hospital.
- No mobile application is provided; the system is web-browser-based only.
- File uploads (e.g., patient X-rays, documents) are not supported in the current implementation.
- Audit logging of user actions is not implemented.
- Password recovery via email requires proper SMTP configuration in the .env file; if unconfigured, reset emails will not be delivered.

4. Target Users

The system is designed for three defined user roles:

Administrator

Full access to all five modules. Responsible for overall system oversight, staff management, billing and financial reports, treatment recording, and system configuration. Administrators can view all dashboards and perform all CRUD operations across the system.

Receptionist

Access to patient registration and update functions, and appointment scheduling. The Receptionist registers new patients, maintains patient demographic data, and schedules outpatient appointments. They do not have access to clinical records, ward management, or billing.

Charge Nurse

Access to medical records, ward and bed management, and admission tracking. The Charge Nurse manages patient admissions and discharges, monitors medication records, assigns beds to patients, and oversees ward occupancy. They do not have access to billing or staff management.

5. Tools and Technologies Used

Category	Technology	Version / Notes
Backend Framework	Laravel	v13.7
Programming Language	PHP	^8.3
Database	PostgreSQL	Relational; stored functions for DML
Frontend Templating	Laravel Blade	Standalone full-HTML templates
Frontend Scripting	Vanilla JavaScript	Dynamic forms, subtotal calculation
Styling	Custom CSS	Per-module CSS files under public/css/
Authentication	Laravel Breeze	Views rewritten to plain HTML
Session / Auth	Laravel Sanctum	4.0
Package Manager (PHP)	Composer	With PSR-4 autoloading
Package Manager (JS)	npm	With Vite for asset bundling
Build Tool	Vite	Via laravel/vite-plugin
Testing	PHPUnit	12.5

Key Architectural Decisions

- All INSERT, UPDATE, and DELETE operations go through PostgreSQL stored functions called via DB::select() — Eloquent ORM is not used for write operations.
- All DDL (function/trigger creation) in migrations uses DB::unprepared() with DROP FUNCTION IF EXISTS guards.
- Middleware aliases are registered in bootstrap/app.php (Laravel 11+ style).

6. System Features

Module 1 — Patient Management

- Patient Registration: Register new patients with full demographic details (name, address, contact, date of birth, next of kin).
- Patient Profile Update: Edit and update existing patient information.
- Medical Records: View a patient's full medical record, including medication history and diagnosis records from Module 4.
- Medication Monitoring: View, add, and update medication records for admitted patients.
- Admission Tracking: Record patient admissions, expected discharge dates, and discharge events.

Module 2 — Staff and Department Management

- Staff Registration: Add new staff members with personal details, staff type, salary, and contracted hours.
- Staff Profiles: View complete staff profiles including qualifications and work experience.
- Qualification & Experience Records: Record professional qualifications and employment history.
- Ward Assignment: Assign staff members to specific wards.
- Staff Schedule/Rota: Create and view shift schedules for staff across wards.

Module 3 — Ward and Bed Management

- Ward Management: Create and maintain ward records (ward name, type, total beds).

- Bed Management: Add beds to wards, view per-bed status (Available/Occupied/Maintenance).
- Bed Assignment: Assign available beds to admitted patients; bed status updates automatically.
- Bed Release: Release beds upon patient discharge, returning them to Available status.
- Bed Availability Dashboard: View real-time summary of occupied vs. available beds per ward.

Module 4 — Appointment and Treatment

- Appointment Scheduling: Schedule outpatient appointments, selecting patient, doctor, date/time, and reason.
- Appointment Management: Cancel or update appointment status (Scheduled, Completed, Cancelled).
- Treatment Recording: Record treatments linked to appointments, including procedures performed and notes.
- Diagnosis Recording: Record diagnosis details, treatment type, and assigned doctor; linked to Module 1 Medical Records.
- Treatment Staff Assignment: Assign clinical staff to treatment records.

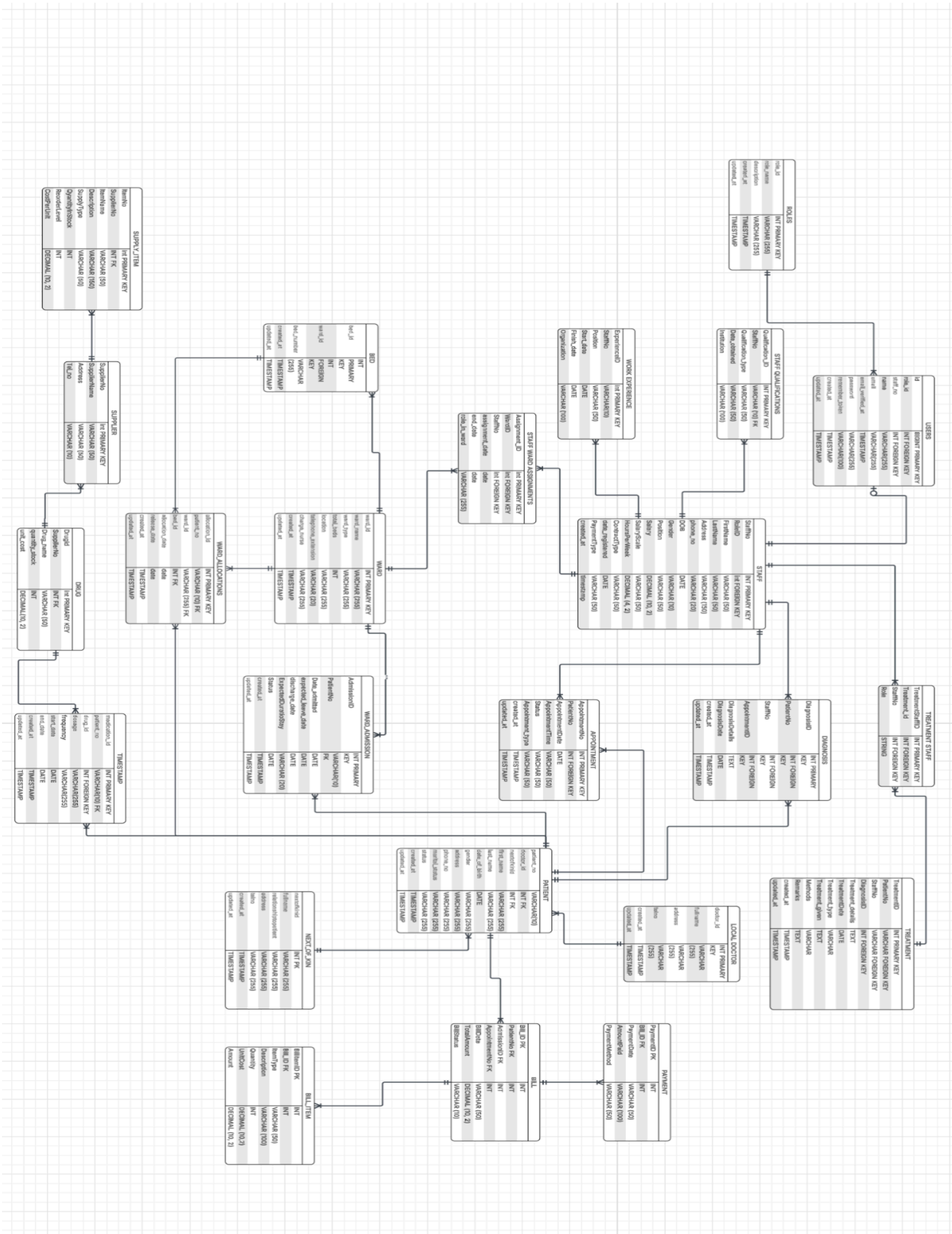
Module 5 — Billing and Reporting

- Bill Generation: Generate itemized bills for patients with multiple line items. Total is computed automatically.
- Bill Detail View: View full bill breakdown with items, payment history, and outstanding balance.
- Payment Recording: Record single or partial payments (Cash, Credit Card, Insurance, Bank Transfer). Bill status updates automatically.
- Bill Cancellation: Cancel unpaid or partially paid bills (enforced by database trigger).
- Financial Reports: Summary dashboard showing total revenue, total bills, paid/unpaid counts, and recent 10 bills.

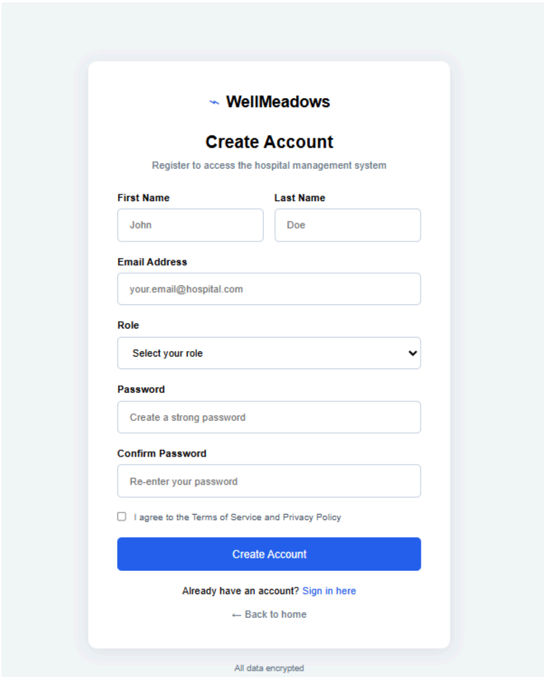
Authentication and RBAC

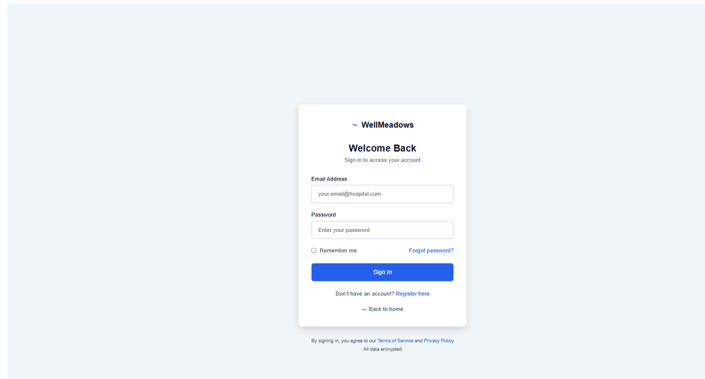
- Secure login with email and password via Laravel Breeze authentication.
- Password reset via email link (requires SMTP configuration).
- Role-based dashboard: users are automatically redirected to their role dashboard after login.
- All routes are protected by auth middleware; role-sensitive routes additionally require the role: middleware.

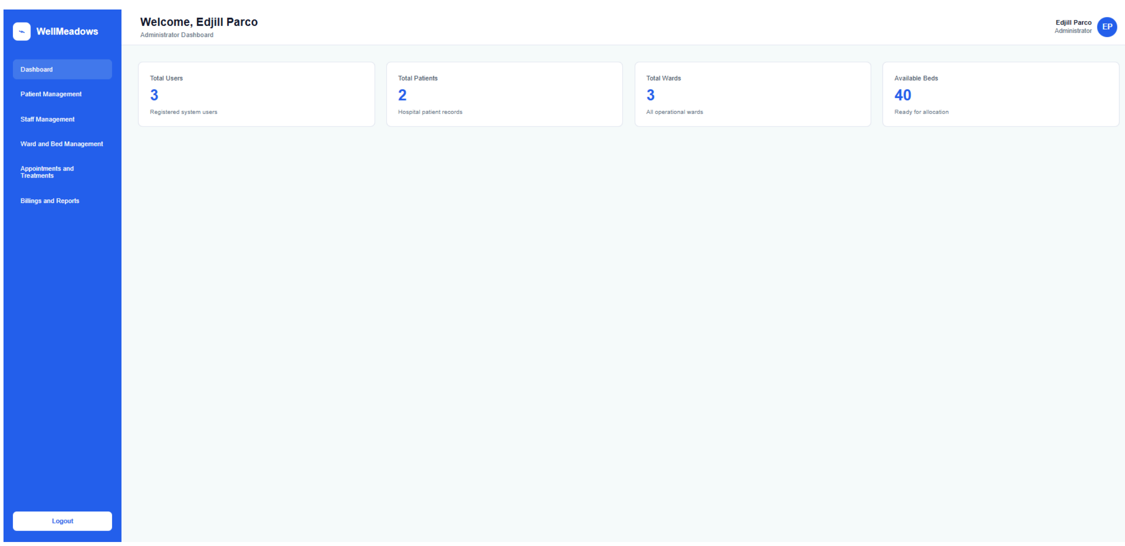
7. DATABASE DESIGN

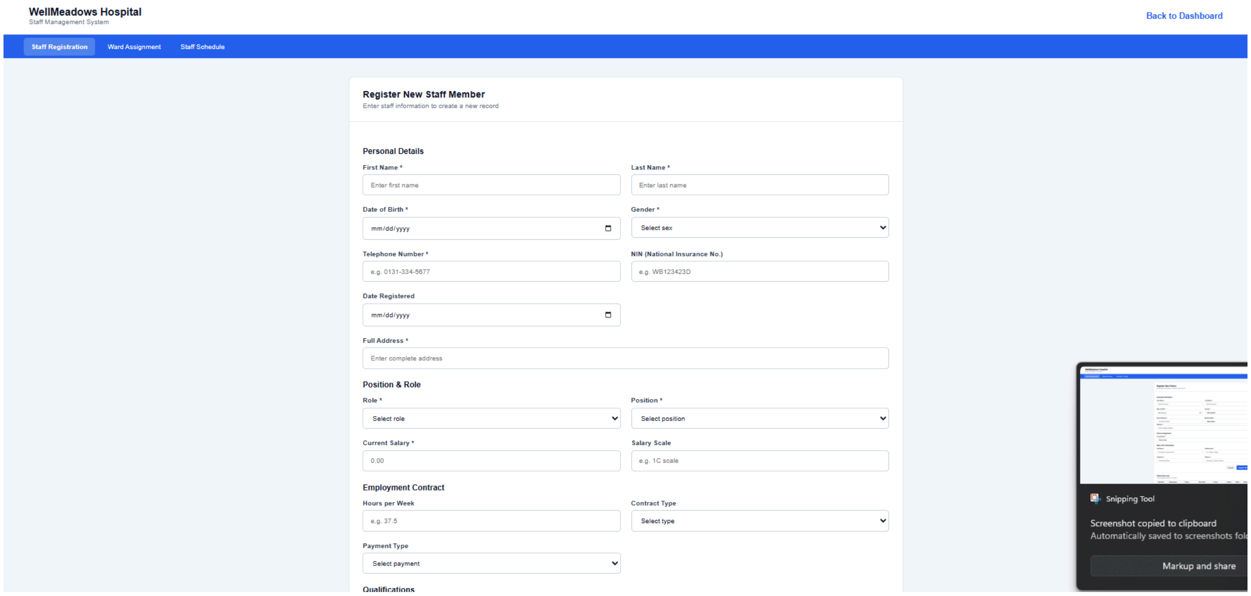


8. SYSTEM SCREENSHOTS









WellMeadows Hospital
Staff Management System

Back to Dashboard

Staff Registration | Ward Assignment | Staff Schedule

Register New Staff Member
Enter staff information to create a new record

Personal Details

First Name * — Enter first name
Last Name * — Enter last name
Date of Birth * — mm/dd/yyyy
Gender * — Select sex
Telephone Number * — e.g. 0131-334-5677
NIN (National Insurance No.) — e.g. WB123423D
Date Registered — mm/dd/yyyy
Full Address * — Enter complete address

Position & Role

Role * — Select role
Position * — Select position
Current Salary * — 0.00
Salary Scale — e.g. 1C scale

Employment Contract

Hours per Week — e.g. 37.5
Contract Type — Select type
Payment Type — Select payment

Qualifications

Snipping Tool
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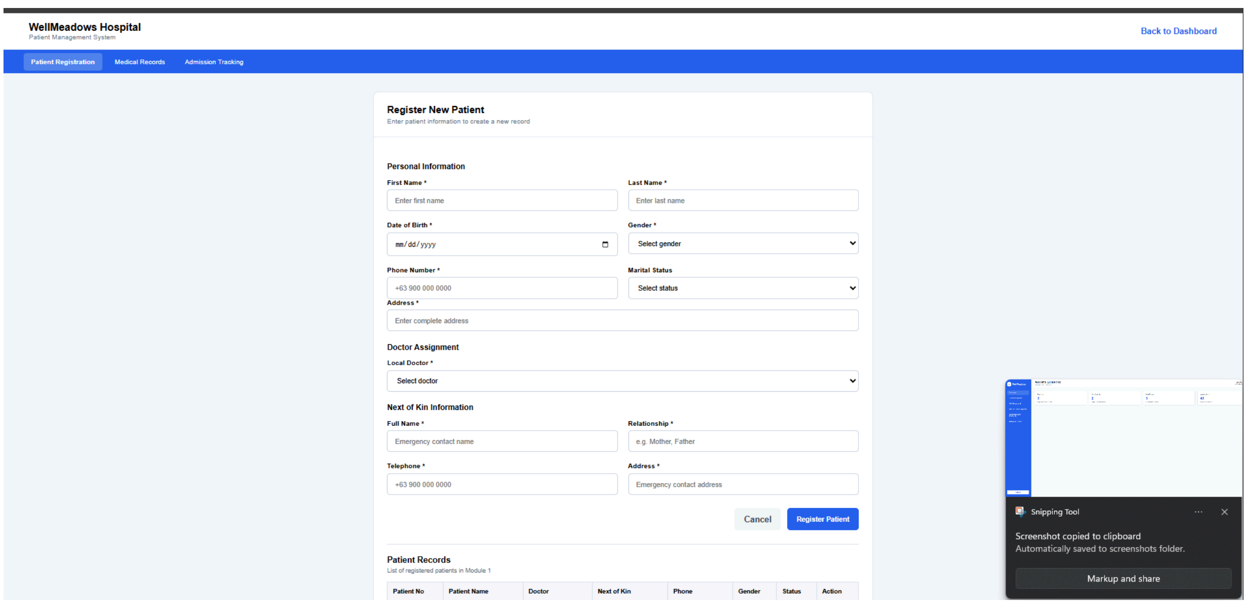
Markup and share



WellMeadows Hospital
Patient Management System

Back to Dashboard

Patient Registration | Medical Records | Admission Tracking

Register New Patient
Enter patient information to create a new record

Personal Information

First Name * — Enter first name
Last Name * — Enter last name
Date of Birth * — mm/dd/yyyy
Gender * — Select gender
Phone Number * — +63 900 000 0000
Marital Status — Select status
Address * — Enter complete address

Doctor Assignment

Local Doctor * — Select doctor

Next of Kin Information

Full Name * — Emergency contact name
Relationship * — e.g. Mother, Father
Telephone * — +63 900 000 0000
Address * — Emergency contact address

Cancel | Register Patient

Patient Records
List of registered patients in Module 1

Patient No	Patient Name	Doctor	Next of Kin	Phone	Gender	Status	Action

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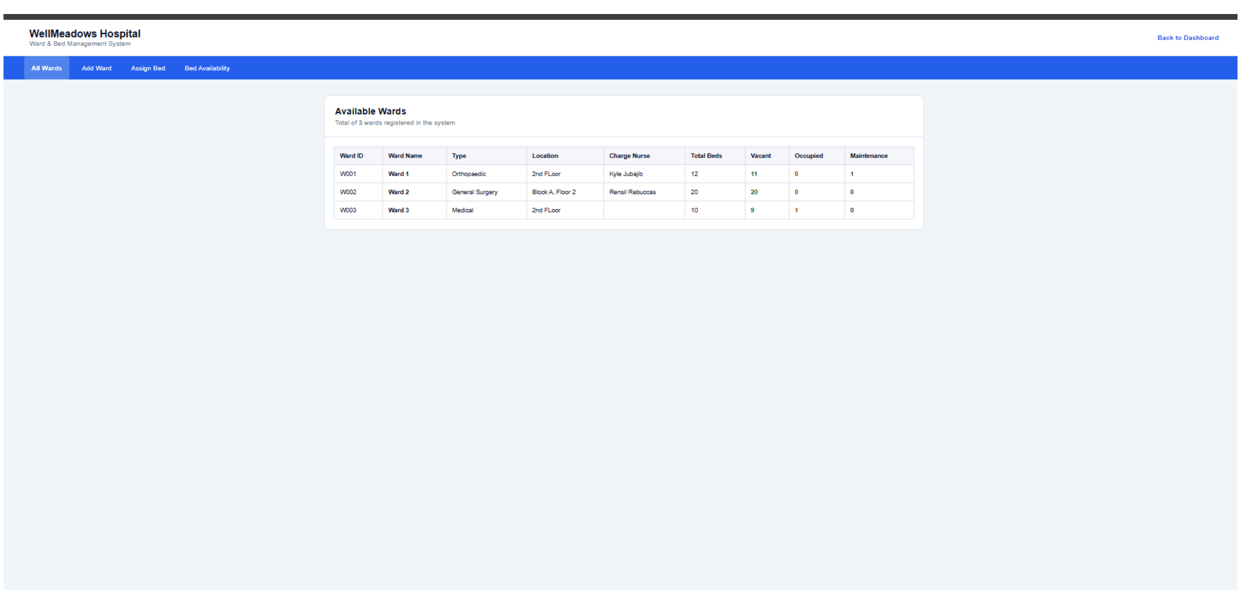


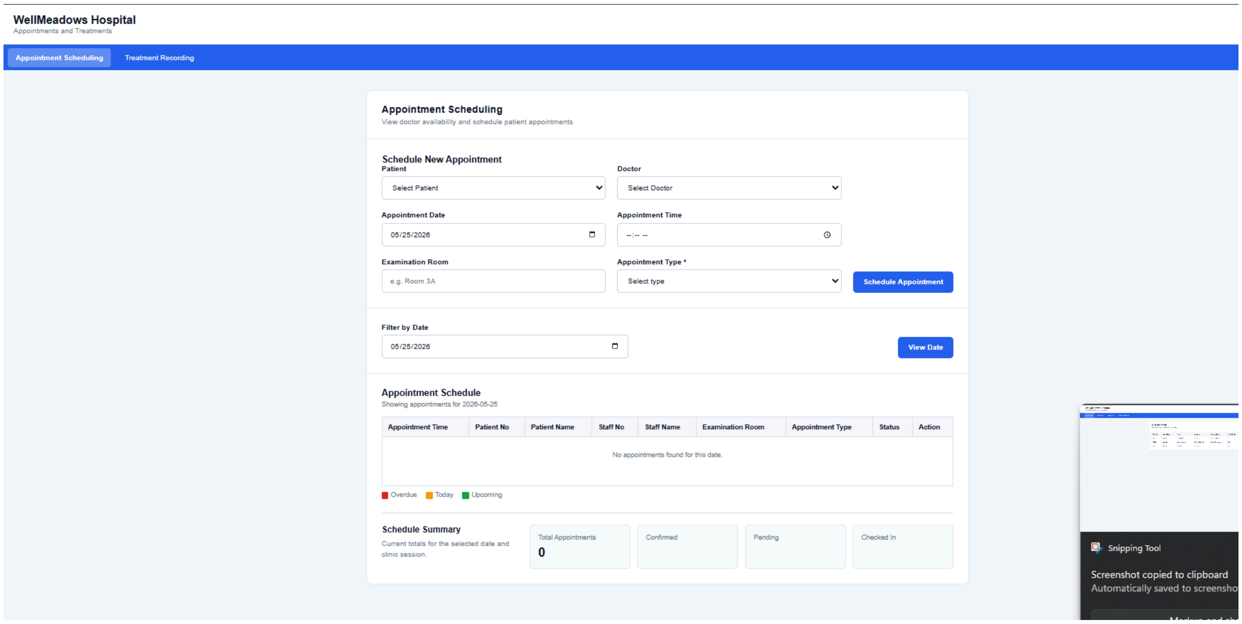
WellMeadows Hospital
Ward & Bed Management System

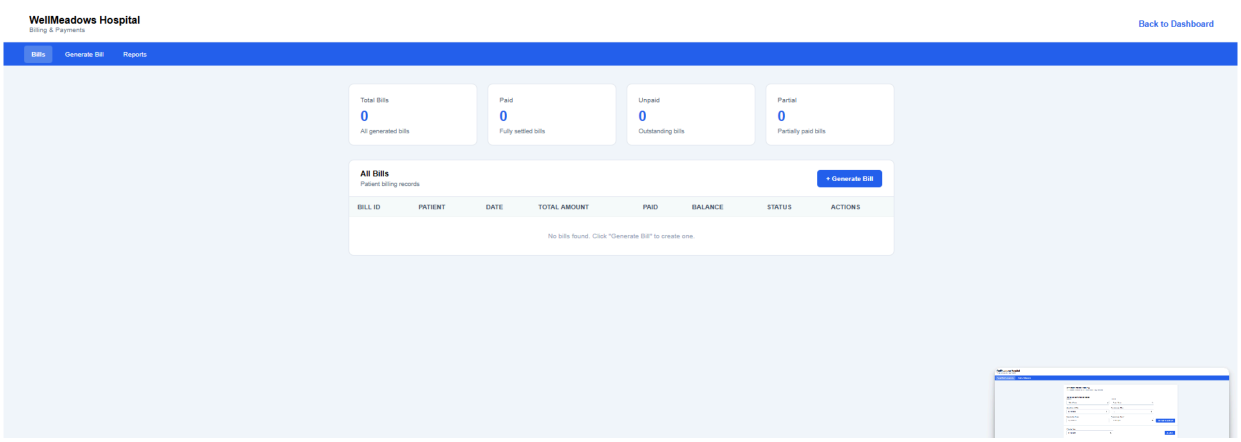
Back to Dashboard

All Wards | Add Ward | Assign Bed | Bed Availability

Available Wards
Total of 3 wards registered in the system

Ward ID	Ward Name	Type	Location	Charge Nurse	Total Beds	Vacant	Occupied	Maintenance
W001	**Ward 1**	Orthopaedic	2nd FLoor	Kyle Jubajib	12	11	0	1
W002	**Ward 2**	General Surgery	Block A, Floor 2	Rensil Rebuccas	20	20	0	0
W003	**Ward 3**	Medical	2nd FLoor		10	9	1	0





9. Source Code Explanation

Directory Structure

Path	Purpose
app/Http/Controllers/	Application logic — one controller per feature/module

Path	Purpose
app/Http/Middleware/RoleMiddleware.php	RBAC enforcement — checks user role against allowed roles
app/Models/	Eloquent models (read-only use for relationships)
bootstrap/app.php	Middleware alias registration (Laravel 11+ style)
database/migrations/	Schema migrations + PostgreSQL function/trigger migrations
public/css/	Per-module CSS files (module1css/, module2css/, etc.)
resources/views/auth/	Login, register, forgot-password, reset-password
resources/views/dashboards/	Role-specific dashboards (admin, receptionist, charge nurse)
resources/views/module1/	Patient management views
resources/views/staff/	Staff management views
resources/views/module3/	Ward and bed management views
resources/views/module4/	Appointment and treatment views
resources/views/module5/	Billing and reporting views
routes/web.php	All application routes organized by module and role

Controllers

Controller	Module	Key Methods
AdminDashboardController	—	index() — fetches counts for summary cards
RoleDashboardController	—	receptionist(), chargeNurse() — role-specific stats

Controller	Module	Key Methods
PatientController	Module 1	create, store, edit, update
MedicationRecordController	Module 1	index, show, store, update — includes diagnosis from Module 4
AdmissionTrackingController	Module 1	index, store, discharge
StaffController	Module 2	index, show, store, edit, update, destroy, wardAssignment, schedule, storeRota
WardBedManagementController	Module 3	index, storeWard, updateWard, storeBed, assignBed, releaseBed, bedAvailability
AppointmentController	Module 4	create, index, store, cancel, updateStatus
TreatmentController	Module 4	create, index, store, edit, update, destroy
TreatmentStaffController	Module 4	Staff assignment to treatment records
BillingController	Module 5	index, create, store, show, recordPayment, storePayment, cancel, reports

PostgreSQL Stored Functions

Module	Functions
Module 1	Fn_register_patient + nextofkin, fn_update_patient, fn_store_medication, fn_update_medication, fn_store_admission, fn_discharge_patient
Module 2	fn_create_staff, fn_update_staff, fn_delete_staff, fn_assign_staff_to_ward, fn_store_rota
Module 3	fn_create_ward, fn_update_ward, fn_delete_ward, fn_add_bed, fn_assign_bed, fn_release_bed, fn_update_bed_status

Module	Functions
Module 4	fn_book_appointment, fn_cancel_appointment, fn_update_appointment_status, fn_record_treatment, fn_record_diagnosis
Module 5	fn_generate_bill, fn_add_bill_item, fn_record_payment, fn_cancel_bill + trigger trg_prevent_payment_on_cancelled_bill

CSS Structure

File/Folder	Used By
public/css/module1css/	Patient registration, medical records, admission tracking pages
public/css/module2css/	Staff management pages (base, registration, profile, schedule, ward assignment)
public/css/module3css/	Ward and bed management pages
public/css/module4css/	Appointment and treatment pages
public/css/module5css/module5.css	Billing and reports pages
public/css/admindash.css	All three role dashboards
public/css/landing.css	Public landing page
public/css/login.css	Login and authentication pages

AI Use Declaration

Tool: Claude, ChatGPT and Gemini — used for debugging, code suggestions, and documentation support during the development of WellMeadows Hospital Management System.

Required Information	**Description**
AI Tool Used	ChatGPT, Claude and Gemini
Purpose of Use	• Debugging PostgreSQL stored function errors • Code suggestions for Laravel controllers and Blade views • Generating SQL fixes for foreign key and constraint issues • Documentation structuring and formatting
Sample Prompts Used	1. "fn_add_medication_record() is receiving drug_id as null — fix the PostgreSQL function" 2. "fn_delete_staff violates foreign key on staff_qualifications — fix with cascade delete" 3. "The Doctor dropdown in Module 4 shows all staff — filter by position = Doctor only"
Output Used	• SQL fixes for fn_add_medication_record and fn_delete_staff were adopted directly after testing • Controller query fix for doctor dropdown filter was adopted with minor adjustments
Student Reflection	Claude and ChatGPT was used primarily as a debugging and code suggestion tool, not as a replacement for understanding. Each suggested fix was reviewed, tested in the browser or psql, and verified before being accepted. The student learned to identify PostgreSQL function overload issues, foreign key cascade strategies, and role-based access control patterns in Laravel through the process of reviewing AI suggestions and confirming results against actual system behavior.

Challenges Encountered

While developing the WellMeadows Hospital Management System, the team encountered several challenges during both the frontend and backend development. One of the main problems was handling database relationships and foreign key errors in PostgreSQL, especially when connecting patient, staff, appointment, and ward records. The team also experienced issues while changing some fields, such as patient_no and staff_no, from integer to varchar format.

Another challenge was implementing role-based access in Laravel. Redirecting users to the correct dashboard and restricting unauthorized access required proper middleware and route configuration. There were also validation issues, duplicate data errors, and trigger conflicts during patient registration and appointment scheduling.

The team solved these problems through continuous testing, debugging, database adjustments, and collaboration using GitHub. Through teamwork and research, the developers were able to improve the system and complete the required functionalities successfully.

Conclusion and Recommendations

The WellMeadows Hospital Management System was successfully developed to help manage hospital operations such as patient registration, staff management, ward and bed assignment, appointment scheduling, treatment recording, and billing. The system helped organize hospital records more efficiently and reduced manual processing through a centralized web-based platform.

The project also helped the developers improve their skills in web development, database management, and system integration using Laravel, PostgreSQL, HTML, CSS, and JavaScript.

For future improvements, the team recommends adding a mobile application using Flutter, improving system security, adding notification features, and enhancing report generation. Additional modules and features may also be added to further improve the overall efficiency and usability of the system.